

Feryadi Yulius

UNDERGRADUATED DATA SCIENCE STUDENT

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Summary

Data Science student at Sumatera Institute of Technology with a passion for technology, mathematics, data science, and machine learning especially in NLP and Image Processing. Fluent both in English and Bahasa Indonesia. Mastering Python, R, and SQL for data analysis. With strong leadership, communication, critical thinking, and analytical skills, envisions making meaningful contributions to data-driven decision-making and technology.

Experience

Data Scientist Research Assistant Intern | Indonesian National Research and Innovation Agency (BRIN-RI) June 2025 – Now

- Drafted a research paper on deep learning modeling for plankton identification from Belitung Island.
- Trained and evaluated four deep learning architectures (AlexNet, DenseNet, VGG16, and InceptionV3) on the Belitung plankton dataset, achieving the highest accuracy of 95% with the DenseNet model.
- Performed manual labeling and segmentation of 86 microscopic plankton images, and applied data augmentation techniques to expand the dataset to over 500 annotated images.
- Developed and refined an Android-based plankton identification application integrated with a mobile microscope.

Statistical for Data Science Practicum Assistant | Sumatera Institute of Technology January 2025 – June 2025

- Taught statistical concepts (hypothesis testing, regression) to 60+ students.
- Assisted in 10 laboratory sessions on R tools.
- Managing 500+ pre-test data and student assignments
- Mentored 1 students group in capstone projects for real-world problems.
- Evaluating students' learning outcomes

Programming Algorithm Practicum Assistant | Sumatera Institute of Technology September 2024 – December 2024

- Delivered practical programming labs to 60+ students to understanding of the Programming Algorithms course
- Assisted in 10 laboratory sessions, helping 95% of students successfully complete Python algorithmic projects.
- Managing 500+ pre-test data and student assignments
- Motivating students
- Evaluating students' learning outcomes

Basic Math Tutorial Assistant | Sumatera Institute of Technology September 2023 – January 2025

- Teaching 10 basic math module to 130+ first-year students across 2 semester.
- Discussing issues raised by the professor, answering questions from mid-semester and end-of-semester exams.
- Designed and assigned 10+ self-study task modules per semester, resulting in an 85% completion rate and improved independent problem-solving skills.
- Increased student enthusiasm through interactive workshops, reflected in a 4.5/5 satisfaction score in post-semester feedback surveys.
- Evaluate the comprehension and difficulties of first-year students to avoid hindering their learning.

Education

Data Science | Sumatera Institute of Technology September 2022 - Now

- CGPA: 3.43/4.00

Data Science Project

Phytoplankton Classification from Belitung Coastal Waters using Deep Learning

August 2025

Processed and cleaned 236 images across 26 phytoplankton classes from Belitung coastal waters. Expanded the dataset to over 10,000 images through 9 augmentation techniques to address class imbalance. Trained and evaluated CNN models across 13 architecture variants including ResNet, DenseNet, EfficientNet, MobileNet, InceptionV3, and ConvNeXt to improve classification performance.

Automatic Phytoplankton Classification Android Application

August 2025

Developed an Android-based mobile application for real-time phytoplankton classification. Implemented using Kotlin and Android SDK with TensorFlow Lite integration for on-device inference. Configured app compatibility with Min SDK 24 (Android 7.0) up to Target SDK 36 (Android 16). Integrated Firebase for database management, enabling scalable storage and retrieval of classification results.

Outfit Recommendation Chatbot using NVIDIA

June 2025

Developed a personalized outfit recommendation system using Retrieval-Augmented Generation (RAG) architecture powered by NVIDIA LLM (Llama-3.1 Nemotron Nano 8B). Integrated FAISS for fast similarity search on vectorized instruction-following dataset. Applied prompt engineering to fine-tune responses in the style of a fashion stylist. The system provides context-aware, text-based outfit suggestions with potential for future multimodal expansion and real-world deployment.

Infographic: Exploring Rice Production in Indonesia

April 2025

Behind Indonesia's glittering achievement as one of the world's largest rice producers lies a paradox that challenges national food security. The infographic "From Fields to Distribution" reveals a stark contrast between impressive production milestones and the unequal realities of distribution. An infographic submitted to the 2025 Gammafest IPB Statistical Visualization Competition.

Toy Store KPI Analysis with PowerBI Dashboard

March 2025

As one of the most dynamic sectors in retail, the toy industry is undergoing rapid changes driven by product innovation and shifting consumer preferences. "Maven Toys," as one of the key players in this industry, must have a deep understanding of its sales performance in order to compete effectively. In this context, the use of BI dashboards is becoming increasingly important to provide data-driven insights to management. Dashboards not only present information visually but also enable in-depth interactive analysis.

Plankton Classification using ResNet

February 2025

Plankton classification model using Deep Learning based on the ResNet-101 architecture pre-trained on ImageNet, with fine-tuning applied to a combined dataset of 3,144 images spanning 19 marine plankton species (sourced from internet images, lecture materials, and the WHOI Plankton Dataset). The model was modified by replacing the final fully connected layer to output 19 classes, trained using the Adam optimizer (learning rate = 0.001) with a StepLR scheduler (reducing the learning rate every 7 epochs), and achieved accuracies of 99.55% (training), 88.17% (validation), and 99.15% (testing).

Spices Detection With Streamlit

November 2024

Spice Detection is an AI-powered online platform designed to identify 31 varieties of Indonesian spices through image analysis. Developed using Python and implemented with the Streamlit framework, this platform boasts a prediction accuracy of up to 83% and aims to simplify the identification and enrich public knowledge of Indonesia's diverse spices.

Know Your Batik Website With Streamlit

October 2024

Know Your Batik is a website that allows users to detect types of batik based on uploaded images or photos. To address the lack of public understanding about the variety of batik and to prevent cultural claims by other countries, the Know Your Batik website serves as an innovative solution. This website is designed to help users recognize and understand different types of batik through image processing technology.

UPN Jatim Data Science National Seminar

September 2024

Conducted a seminar and publication with the title paper "Variable Selection Methods in Linear Regression Modeling of Rainfall Data in Lampung Province."

Journal Mancing With Streamlit

November 2023

A final project for a Programming Algorithms course involved creating a fishing log app using Python libraries like Streamlit, Sqlite3, and Pandas. The application, which has been deployed on the Streamlit community and stored in a GitHub repository, functions similarly to a note-taking app on mobile devices but is tailored for fishing activities. This specialized app offers various features to aid anglers in recording and monitoring their fishing endeavors, aiming to enhance their fishing experiences.

Certification

Data Scientist Associated

Credential ID: [DSA0014978540656](#)

October 2024 – October 2026

Python Data Associated

Credential ID: [PDA0015038237495](#)

September 2024 – September 2026

Technical Skills

Programming Language : Python, R, SQL

Dashboard Tools : PowerBI, Tableau, Streamlit

Tools : MS Excel, MS Words, Google Colab, Jupyter Notebook

Database : MySQL, PostgreSQL, Hadoop